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Do Alliances Deter War?

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This research note reexamines the ongoing debate about whether security alliances help prevent conflicts. It specifically reevaluates recent scholarly exchanges between deterrence and Steps-to-War studies. Deterrence scholars argue that alliances reduce the likelihood of conflict initiation, while Steps-to-War scholars contend that forming an alliance can actually incite conflict. Both sides provide evidence to support their theories by analyzing the effectiveness of defensive pacts in deterring or failing to deter militarized interstate disputes. However, recent studies have overlooked testing and reporting whether alliances prevent the most destructive form of conflict: *war*. This omission is surprising, given that the primary goal of alliances is to prevent *war*. This research note conducts a canonical directed dyad-year analysis of 1,077,992 observations from 1816 to 2000 and finds compelling evidence in support of deterrence theory, but not for the Steps-to-War theory. As a result, this empirical research clarifies the debate between these two leading theories in international relations.

Esta nota de investigación reexamina el actual debate con respecto a si las alianzas en materia de seguridad ayudan a prevenir conflictos. En concreto, esta nota reevalúa los intercambios académicos recientes que han tenido lugar entre los estudios en materia de disuasión y los estudios referentes a los pasos que conducen hacia la guerra. Los académicos del ámbito de la disuasión argumentan que las alianzas reducen la probabilidad de que se inicie un conflicto, mientras que los académicos que se especializan en los pasos que conducen hacia la guerra argumentan que formar una alianza puede, en realidad, incitar al conflicto. Ambas partes aportan pruebas que respaldan sus respectivas teorías y analizan la eficacia que tienen los pactos defensivos para disuadir o no las disputas interestatales militarizadas. Sin embargo, los estudios recientes tienden a pasar por alto la importancia que tiene comprobar y divulgar si las alianzas ayudan a prevenir la forma más destructiva de conflicto: la guerra. Esta omisión no deja de ser sorprendente, teniendo en cuenta que el objetivo principal de las alianzas es prevenir la guerra. Esta nota de investigación realiza un análisis diádico canónico dirigido de 1 077 992 observaciones, desde el año 1816 hasta el año 2000, y encuentra evidencia convincente que apoya la teoría de la disuasión, pero no la teoría de los pasos que conducen hacia la guerra. En consecuencia, esta investigación empírica aclara el debate entre estas dos principales teorías de las relaciones internacionales.

Cette note de recherche réexamine le débat actuel sur la question suivante : les alliances de sécurité permettent-elles de prévenir les conflits ? Plus précisément, elle réévalue les échanges académiques récents entre les études de la dissuasion et des étapes qui mènent à la guerre. Les chercheurs spécialisés dans la dissuasion affirment que les alliances ré-

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duisent la probabilité de déclenchement d'un conflit tandis que ceux qui s'intéressent aux étapes qui mènent à la guerre soutiennent que la constitution d'une alliance peut en fait inciter au conflit. Les deux camps présentent des éléments probants pour étayer leur théorie en analysant l'efficacité des pactes de défense quand il s'agit de prévenir ou d'échouer à prévenir les litiges interétatiques militarisés. Néanmoins, de récentes études omettent de tester et d'indiquer si les alliances préviennent la forme la plus destructrice de conflit : la guerre. Cette omission paraît surprenante, car l'objectif premier d'une alliance est d'empêcher la guerre. Cette note de recherche conduit une analyse par année de dyade dirigée et canonique de 1 077 992 observations entre 1816 et 2000. Elle trouve des éléments qui viennent fortement étayer la théorie de dissuasion, mais pas celle des étapes qui mènent à la guerre. Par conséquent, cette recherche empirique éclaircit le débat entre ces deux théories dominantes en relations internationales.

Introduction

This study reexamines a long-standing debate regarding whether security alliances help prevent conflicts. There is no universally accepted argument about the role of alliances within the deterrence literature. Scholars provide various theoretical explanations and empirical findings concerning the effectiveness of security alliances in deterring conflicts (Levy 1981). Some researchers argue that these alliances can help prevent conflicts (e.g., Levy 1981; Leeds 2003; Quackenbush 2010; Johnson and Leeds 2011), while others contend that alliances do not lead to peace (e.g., Huth and Russett 1984; Vasquez 1993; Snyder 1997; Fearon 2002; Senese and Vasquez 2008; Kenwick, Vasquez, and Powers 2015). Additionally, some studies demonstrate an ambiguous relationship between alliances and conflicts (e.g., Smith 1995).

The debate about the relationship between alliances and war is central to the existing literature,[1] as one of the primary goals of two states entering into a formal alliance is to prevent or avoid *war*. As Levy (1981, 581) notes, "alliances have often been formed in response to unstable conditions, rising tensions, and the anticipation of a probable *war*." Furthermore, Morrow (2000, 63) states that "two or more states form an alliance when they conclude a treaty that obliges them both to take certain actions in the event of *war*."

Leeds (2015, 6–8) dedicates a subsection to exploring the relationship between alliances and war, a major question in alliance politics. In her review of key studies, including those by Pressman (2008), Senese and Vasquez (2008), Johnson and Leeds (2011), and Yuen (2009), she concludes that "there remains some disagreement about the overall effect of alliances on conflict" (p. 1). Consequently, scholars continue to conduct empirical evaluations to address these disagreements.

Not long ago, a prominent journal featured an engaging debate on the connection between alliances and war, involving experts in deterrence and the Steps-to-War framework. This discussion, referred to as the Scholarly Dialogue in the *Journal of Politics*, primarily focused on examining which theories were supported by statistical evidence rather than presenting new theoretical innovations.[2]

[1] "The study of alliances has traditionally focused on states and war, with alliances being a tool with which the former could manage the latter" (Rynning and Schmitt 2018, 653).

[2] Several studies have investigated whether alliances can serve as a deterrent and have examined the variation in their effectiveness. For example, Clare (2013) studies whether democratic allies are more effective at deterrence than non-democratic ones. Johnson, Leeds, and Wu (2015) explore whether alliances with greater capability or credibility are more successful in deterring aggression. Bak (2018)considers whether the proximity of allies enhances deterrent effects. Chen (2021) analyzes whether trade relationships with a state's allies improve deterrence (for a dissenting view,

The Scholarly Dialogue emphasizes the impact of alliances on militarized interstate disputes (MIDs), many of which were accidental, and does not prioritize war as the main outcome. This focus is misplaced, as scholars in international relations have maintained that nations establish security alliances to prevent *war* (e.g., Levy 1981; Morrow 2000). This study points out that the Dialogue fails to test and report on the effects of alliances on war. By centering on war, this empirical research further contributes to the ongoing debate regarding the relationship between alliances and conflict.

This study aims to reintroduce war into empirical models to reevaluate which theory provides greater explanatory power: deterrence or Steps-to-War. This reevaluation is crucial because understanding whether alliances have a pacifying or provocative effect is a key research question with significant policy implications. Analyzing full-scale warfare in the context of alliances is a valuable contribution, as it seeks to shed light on the fundamental theoretical expectations of alliance politics. As Kenwick and Vasquez (2017, 331) point out, studying war has important implications: "From a policy perspective, what is the point of deterring MIDs if the same logic cannot prevent war?"

In a word, the main contribution of this research note is to address the gap in the debate between two prominent groups of international relations scholars by providing evidence that the Steps-to-War theory has limited explanatory power when determining war outcome, while deterrence theory elucidates the pacifying effect of alliances.

In the following section, this study offers a concise overview of the scholarly dialogue between researchers of deterrence and those who advocate for the Steps-to-War approach, explores the history of this dialogue in greater detail, and discusses the implications of the main findings.

The Debate Between Deterrence and Steps-to-War Scholars[3]

With a special focus on three stimulating studies (Johnson and Leeds 2011; Kenwick, Vasquez, and Powers 2015; Kenwick and Vasquez 2017), Table 1 summarizes the main contentions between deterrence and Steps-to-War researchers. It demonstrates how the choices of different research designs can lead to different findings.

In their 2015 study, Kenwick, Vasquez, and Powers conducted a matching analysis to demonstrate that alliances can lead to MIDs and wars. To avoid sampling bias, they used the same sample of defense pacts[4] that Johnson and Leeds (2011) had collected for their logistical regression analysis. However, by matching each treated unit with a non-treated unit of similar characteristics, Kenwick, Vasquez, and Powers only utilized 44,929 (4 percent) out of the 1,077,992 observations Johnson and Leeds collected for their logit estimation. This significantly reduced sample size raised concerns from Leeds and Johnson (2017), who pointed out the potential incompatibility of empirical results and estimation methods.

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see Choi 2024). Additionally, Benson and Smith (2023) develop a formal theory explaining why the anticipation of alliance formation or expansion can provoke hostility from adversaries in some instances and not in others. Despite the theoretical advancements and improved research designs provided by these scholars, their findings are not directly relevant to the issue addressed in this research note. In their response, Kenwick and Vasquez (2017, 331) challenged the work of Leeds and Johnson (2017), stating, "Neither do Leeds and Johnson (2017) address our findings on war," as studied in Kenwick, Vasquez, and Powers (2015). This research note responds to Kenwick and Vasquez's critique as an independent third party and finds that their Steps-to-War theory lacks predictive power regarding the occurrence of war.

[3]This study does not aim to explain the strategic dynamics of the alliance-conflict relationship involving third parties, but rather to review the debate on whether alliances between two states in a dyad deter war.

[4]"Defense pacts are a particular type of military alliance that specifically commit members to assist one another with military force in the event of [an] attack on the sovereignty or territorial integrity of a member" (Leeds and Johnson 2017, 336).

Table 1. Differences in research designs.

	Kenwick, Vasquez, and Powers (2015)	Johnson and Leeds (2011)	Kenwick and Vasquez (2017)
Source	Figure 1 on Page 951	Table 1 on Pages 56 and 61	Figure 2 on Page 333
Main estimation method	Matching	Logistic regression	Logistic regression
Dependent variable	MIDs and war	MIDs and war	MIDs and fatal MIDs
Independent variable: alliances	Alliances are disaggregated into two categories: (1) Target has a defense alliance from birth. (2) Target has a defense alliance formed after birth	Alliances are not disaggregated	Alliances are disaggregated into two categories: (1) Target has a defense alliance from birth. (2) Target has a defense alliance formed after birth
Number of observations	44,929	1,077,992	1,077,992

In response to criticism from Kenwick and Vasquez (2017), Leeds and Johnson (2017) wrote a reply stating that they chose to employ the same research design as Johnson and Leeds (2011), conducting logistical regression on their directed dyad-year observations of 1,077,992. Kenwick and Vasquez's response purported to create a level playing field for researchers with differing views on the role of alliances.

However, Kenwick and Vasquez's (2017) empirical analysis, without explanations, departed from Johnson and Leeds (2011) and even their original study, Kenwick, Vasquez, and Powers (2015), where the dependent variables were MIDs and war. Kenwick and Vasquez reported that alliances led to MIDs and fatal MIDs, but they did not present test results on the effect of alliances on war. The following sections demonstrate that alliances do not prompt war, which challenges the conclusions drawn by Kenwick and Vasquez.

To provide more context, this study gives a detailed background of the Scholarly Dialogue about deterrence and Steps-to-War studies. Scholars such as Leeds (2003), Johnson and Leeds (2011), and Wright and Rider (2014) theorize that forming alliances is a strategy for a threatened state to increase its power and establish a balance of power, which in turn reduces the likelihood of war. They performed regression analysis and uncovered strong evidence for their theoretical argument. However, Kenwick, Vasquez, and Powers (2015) challenge the premise of the deterrence theory. They argue that alliances do not always lead to peace and that their Steps-to-War theory predicts the negative consequences of alliance politics. The Steps-to-War theory suggests that forming a defensive alliance heightens the risk of war by creating security dilemmas. To test their theoretical expectation, Kenwick, Vasquez, and Powers (2015) employed matching analysis, a method different from the standard estimation method, logistic regression, used by previous alliance scholars.

In their response to Kenwick, Vasquez, and Powers (2015), Leeds and Johnson (2017) illuminate differences in research design. They argue that their regression analysis is suitable for studying the deterrent effect of defense pacts, while the matching analysis conducted by Kenwick, Vasquez, and Powers may be more appropriate for testing the Steps-to-War theory. Leeds and Johnson's primary contention is that Kenwick, Vasquez, and Powers' different research designs led them to find evidence against their deterrence theory. As noted, Kenwick, Vasquez, and Powers showed that alliance formation will likely incite MIDs and war. Dafoe, Oneal, and Russett (2013, 202) have suggested earlier that "analyses should be based wherever possible on standard models to minimize the chance that unrelated aspects of the analysis are driving the results." Leeds and Johnson's argument is quite persuasive in light of this suggestion.

Embracing Leeds and Johnson's critiques, Kenwick and Vasquez (2017) replicated their results using the same regression analysis performed by Johnson and Leeds (2011), who produced evidence of deterrence success. Kenwick and Vasquez contend that their replication of Johnson and Leeds' regression models shows "additional support for [their] primary conclusion that alliance formation often fails to deter adversaries and instead increases [the] likelihood of militarized conflict" (p. 329).

However, while re-testing the Steps-to-War theory within the framework of Johnson and Leeds' (2011) research design, Kenwick and Vasquez (2017) focused only on the relationship between alliance formation and (fatal) MIDs. This reexamination is incomplete and inadequate.

First, the reexamination is incomplete because Kenwick and Vasquez did not investigate whether alliances deter *war*, failing to test the most crucial aspect of their Steps-to-*War* theory. Kenwick and Vasquez should have tested the war variable, as they stated that to be on level ground, "[they replicate their] findings using the [same research] design employed by Johnson and Leeds (2011)" (p. 331). This replication analysis is necessary to resolve the differences in the two studies' research design: Johnson and Leeds' (2011, pp. 60-62) logit models included war as an outcome variable in the statistical analyses that used a total observation of 1,077,992, and Kenwick and Vasquez's (2017) regression models did not. It is also important to note that Kenwick and Vasquez deviate even from their original study—Kenwick, Vasquez, and Powers (2015)—where the outcome variables include war in performing a matching analysis with a sample of 44,929, only about 4 percent of Johnson and Leeds' (2011).[5] By including militarized disputes but omitting war, Kenwick and Vasquez (2017, 331) contradicted their own rhetorical question: "From a policy perspective, what is the point of deterring MIDs if the same logic cannot prevent war?" For Kenwick and Vasquez's (2017) Steps-to-*War* theory to supersede Leeds and Johnson's deterrence theory, it must demonstrate that using the same research design, the recent formation of defensive alliances has increased the risk of war.

Second, the reexamination is inadequate because Kenwick and Vasquez's statistical models do not test the critical steps specified in the *Steps*-to-War theory. Their theory suggests that forming an alliance with a third state can threaten the security of another state, prompting the latter to seek a counter-alliance, ultimately leading to an interstate war. The theory's strength lies in its emphasis on the escalating steps—"responses by the challenger in explaining how defensive alliance become linked to [the] onset of war" (Kenwick and Vasquez 2017, 330). However, their statistical models fail to consider the steps to escalate from low-intensity conflict to war. This is surprising given that Johnson and Leeds' (2011, 60-62) research design, which they replicated, already provided the basis for testing the escalation steps from low-intensity conflict to war. This study addresses the shortcomings of Kenwick and Vasquez's analyses by introducing Heckman's selection model, which helps test the logic of the Steps-to-War theory more accurately.

In sum, Kenwick and Vasquez (2017) criticize Johnson and Leeds (2011) and Leeds' (2003) deterrence theory for being too optimistic about the role of alliances in keeping peace between states. They propose the Steps-to-War theory as a more realistic representation of the conflict process, suggesting that alliances can actually worsen conflicts. However, their study only examines militarized disputes and does not consider war, which nations try to prevent by forming security alliances. For the Steps-to-War theory to succeed, it must be validated against war.

[5] Leeds and Johnson (2017, 335) "[did] not believe that [Kenwick, Vasquez, and Powers' (2015)] research design is appropriate for testing the theoretical argument [they] have made in [their] past work," so they seemed to doubt that Kenwick, Vasquez, and Powers' (2015) analysis of the impact of alliances on war is valid.

The following section presents a "careful and scientific empirical analysis" using a canonical directed dyadic dataset from 1816 to 2000 (Kenwick and Vasquez 2017, 333). This study finds no evidence supporting the Steps-to-War theory, but instead, it discovers that Johnson and Leeds' (2011) deterrence theory has significant explanatory power on interstate war. This study provides more relevant statistical testing to clarify the debate between the two contrasting groups of scholars in the areas of defensive alliances and interstate war.

Scrutinizing Kenwick and Vasquez's (2017) Regression Model

This study examines the three models developed by Kenwick and Vasquez, as shown in Online Appendix Table 1. These models form the foundation for the graphical analysis presented in figure 2, which illustrates the main findings of their research. The study period spans from 1816 to 2000, with each model representing a different timeframe. The first model encompasses the entire study period, the second analyzes the years from 1816 to 1945, and the third focuses on the post-1945 period, specifically from 1946 to 2000.

Adopted from Johnson and Leeds' (2011) study, Kenwick and Vasquez build a statistical model as follows:

$$\text{Militarized interstate disputes}_{ijt} = f(\text{target has a defense alliance from birth}_{ijt-1}, \text{target has a defense alliance formed after birth}_{ijt-1}, \text{control variables}_{ijt-1}).$$

Dependent Variable

The term, MIDs refers to all instances where one state threatened, displayed, or used force against another. The dependent variable is dichotomous, coded as "1" when the challenger in a directed dyad initiated a MID of any severity against the target state.[6] Since the dependent variable is dichotomous, Kenwick and Vasquez choose logistic regression as an estimation method. The statistical model accounts for temporal dependence with the cubic polynomial approximation: t, t^2, and t^3.

Two Main Independent Variables

To operationalize two opposing theoretical expectations on the relationship between alliances and militarized disputes, Kenwick and Vasquez re-categorize all defensive alliance observations in Johnson and Leeds' (2011) dataset into two groups: (1) those that have been in place since a directed dyad first entered the data and (2) alliances that were formed afterward.

The first independent variable is labeled "Target has a Defense Alliance from Birth."[7] This variable allows Kenwick and Vasquez to test the validity of Johnson and Leeds' deterrence theory. This variable is expected to reduce the onset of MIDs since defensive alliances should deter militarized disputes successfully.

The second independent variable is "Target has a Defense Alliance formed after Birth."[8] Kenwick and Vasquez create this variable to test their Steps-to-War prediction. This variable is projected to increase the onset of MIDs since defensive alliance formation should be associated with deterrence failure more often than deterrence success.

[6] There are 2,354 MIDs (0.22 percent) among 1,077,992 observations.

[7] There are 234,885 birth alliances (22 percent) among 1,077,992 observations.

[8] There are 394,077 after-birth alliances (37 percent) among 1,077,992 observations.

Control Variables

Six additional variables are included as controls that may influence the main predictors and the outcome variable: Potential challenger has a relevant offensive alliance, Potential challenger has a relevant neutrality pack, Logged distance, Capabilities ratio, Joint democracy, and Similarity in alliance portfolios. Since Johnson and Leeds (2011) explain these controls in detail, this study does not discuss their causal relationship with the outcome variable to save space.[9]

Reproduction

The study successfully reproduces Kenwick and Vasquez's estimates using the replication materials provided on Dataverse.[10] Table 2 displays the replicated estimates across three different periods (Models 1, 2, and 3), and the coefficients' magnitude and sign coincide with those reported by Kenwick and Vasquez. Upon perusing their estimated results, Kenwick and Vasquez conclude that "alliances sometimes reduce the likelihood of MID initiation but that, in the majority of cases (i.e., alliances formed after independence), they have the opposite effect" (p. 21). However, this study finds their conclusion inaccurate, as the results indicate that both deterrence and Steps-to-War theories gain empirical support regardless of the study period.[11] "Target has a defense alliance from birth" is statistically significant across the models and in the expected direction. This lends credence to Johnson and Leeds' deterrence theory, predicting a decrease in militarized disputes. This finding requires revising Kenwick and Vasquez's interpretation: "alliances [not] sometimes [but on average] reduce the likelihood of MID initiation."

This study transforms the coefficients in Models 1–3 into standardized coefficients in Models 4 to 6 to assess the strength and direction of the relationship between variables. This transformation aims to compare the relative importance of deterrence and Stepts-to-War theories. The standardized coefficients show that Johnson and Leeds' deterrence theory has approximately two to three times more explanatory power than Kenwick and Vasquez's Stepts-to-War theory. This suggests that Kenwick and Vasquez's assertion that "alliances sometimes reduce the likelihood of MID initiation but that, in the majority of cases (i.e., alliances formed after independence), they have the opposite effect" is overstated, as the impact of the alliance cases formed from birth is much greater than that of the other alliance cases.

Replication

To test the two opposing theories in the context of war, this study modifies Kenwick and Vasquez's model specification by employing the dependent variable of interstate war in place of MIDs. An interstate war is a conflict that caused at least 1000 battle deaths. This study gathers the war variable from Kenwick and Vasquez's dataset, initially drawn from the Zeev Maoz dyadic data collection. The war variable is dichotomous, with "1" representing the challenger initiating a new war against the target state in a given dyad year and "0" otherwise.[12]

[9]Online Appendix 1 provides a correlation matrix of variables.

[10]See https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/WBWWUO.

[11]It is interesting that Lee (2023) finds that the post-1945 international system significantly increased the number of weaker alliances.

[12]There are 93 wars (0.1 percent) among 1,077,992 observations. Online Appendix 2 shows a correlation matrix of variables. It is worth noting that birth alliance and after-birth alliance variables *negatively* correlate with war.

Table 2. Defense alliance and international conflict.

Variable	Logit								
	All MIDs (replicated)			All MIDs (standardized coeffs)			Wars		
	1816–2000 Model 1	1816–1945 Model 2	1946–2000 Model 3	1816–2000 Model 4	1816–1945 Model 5	1946–2000 Model 6	1816–2000 Model 7	1816–1945 Model 8	1946–2000 Model 9
Target has a defense alliance from birth	−1.327***	−2.216***	−1.112***	−0.2451	−0.179	−0.2137	−2.063***	no	−1.417*
	(0.149)	(0.483)	(0.170)				(0.613)	Variation	(0.731)
Target has a defense alliance formed after birth	0.346***	0.230**	0.657***	0.0746	0.0362	0.1416	−1.308***	−0.476	−0.573
	(0.087)	(0.113)	(0.140)				(0.296)	(0.421)	(0.500)
Potential challenger has a relevant offensive alliance	0.628***	1.297***	−0.122	0.048	0.119	−0.0088	0.954**	1.124***	No
	(0.105)	(0.127)	(0.231)				(0.372)	(0.430)	Variation
Potential challenger has a relevant neutrality pack	0.877***	0.360**	1.145***	0.0859	0.0401	0.108	−0.034	0.300	No
	(0.146)	(0.176)	(0.174)				(0.485)	(0.488)	Variation
Logged distance	−0.971***	−0.551***	−1.187***	−0.3667	−0.2796	−0.4041	−1.091***	−0.769***	−1.241***
	(0.042)	(0.042)	(0.083)				(0.086)	(0.120)	(0.126)
Capabilities ratio	0.324**	0.595***	0.148	0.0506	0.1037	0.0225	0.711*	1.173***	−0.365
	(0.134)	(0.186)	(0.187)				(0.398)	(0.450)	(0.738)
Joint democracy	−0.402***	−0.992***	−0.095	−0.0567	−0.1072	−0.0138	−2.125***	no	−0.596
	(0.104)	(0.238)	(0.124)				(0.754)	Variation	(0.819)
Similarity in alliance portfolios	−0.971***	−0.061	−2.220***	−0.081	−0.0048	−0.1845	−0.796	0.310	−1.265
	(0.237)	(0.209)	(0.481)				(0.601)	(0.814)	(1.667)
Constant	3.322***	−0.476	5.796***				0.192	−3.548***	1.878
	(0.375)	(0.335)	(0.914)				(0.876)	(1.135)	(2.192)
N	1,077,992	211,128	866,864	1,077,992	211,128	866,864	1,077,992	211,128	866,864

Notes: Numbers in parentheses are robust standard errors adjusted for clustering on directed dyads. $^{\text{Variation}}P < .10$; $^{**}P < .05$; $^{***}P < .01$. t , t^2, and t^3 do not appear to economize space.

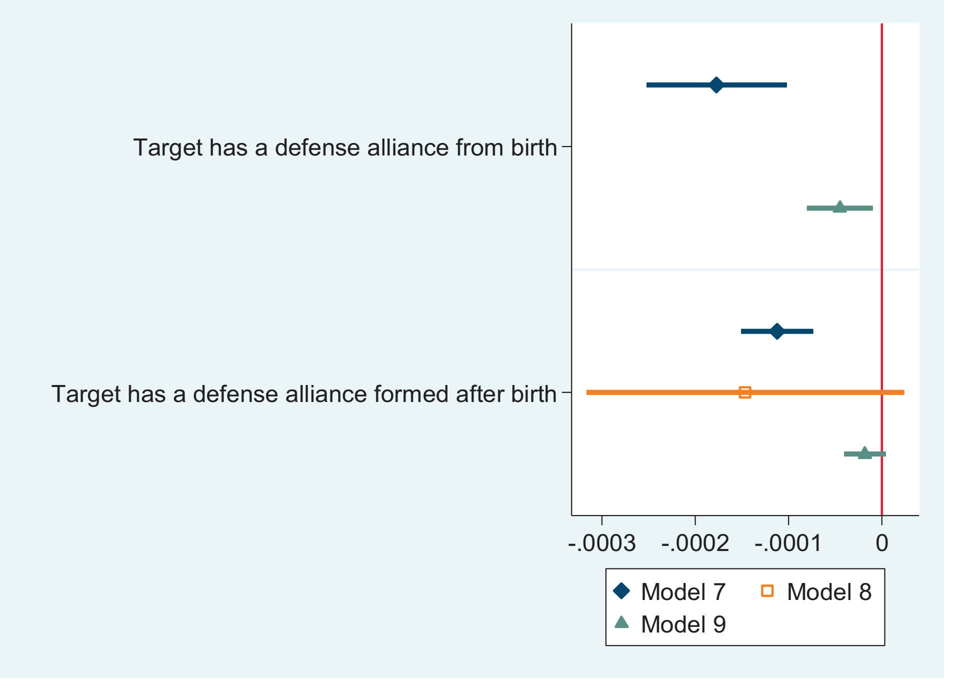

Figure 1. Average substantive effects

Models 7, 8, and 9 in Table 2 display the modified analysis.[13] The results corroborate Johnson and Leeds' (2011) deterrence theory but not the Steps-to-War theory.

Models 7 and 9 suggest that defense pacts, denoted as "Target has a defense alliance from birth," can reduce the likelihood of war, as predicted in deterrence theory. Since the deterrence variable does not vary against the outcome variable during the pre-1946 period, this study obtains no coefficient and standard errors for the variable under Model 8.

Although the coefficient on the Steps-to-War variable (i.e., "Target has a defense alliance formed after birth") in Model 7 is statistically significant, its sign is negative. This means that forming an alliance does not provoke war but lowers the risk of the most destructive conflict. This finding is in stark contrast to the Steps-to-War theory. In addition, the Steps-to-War variable fails to achieve significance in Models 8 and 9.

As the number of observations increases, the statistical significance may not always have practical relevance. To address this concern, this study examines the average substantive effects of deterrence and Steps-to-War variables included in Models 7, 8, and 9. Average substantive effects provide insights into the effect on the probability, which falls within the range of 0–1. They represent the average change—magnitude—in probability when each independent variable increases by one unit. As shown in figure 1, where the length of each bar represents the level of the magnitude, neither well established nor recently formed alliances significantly *increase* the probability of interstate war.

[13]It is interesting to examine how the model fits the data. This study employs Hosmer and Lemeshow's goodness-of-fit test. With a *p*-value of 0.26, this study can say that Hosmer and Lemeshow's goodness-of-fit test indicates that Model 7 fits the data well; Models 8 and 9 also fit the data well, as the *p*-values are 0.66 and 0.38, respectively.This study also tests whether including independent variables significantly increases the area under the Receiver Operating Characteristic (ROC) curve. The area under the ROC curve is a summary metric that quantifies the model's overall performance, with a higher AUC indicating better performance. All three models perform well, as the area under the ROC curve is 0.88 for Model 7, 0.79 for Model 8, and 0.92 for Model 9.

Dealing with an Excessive Number of Irrelevant Zeros

Although wars are often portrayed in the mass media, they are actually quite rare occurrences in international relations. Most disputes between states are resolved without resorting to violence. As a result, war data tends to contain many irrelevant zeroes, and statistical models should consider this. This study utilizes two methods: rare event logit and politically relevant dyads (for applications, see Choi 2013, 2022). Table 3 presents the estimated results from these two methods, which align with Models 7 to 9 in Table 2. While deterrence theory is supported by evidence demonstrating the pacifying effect of alliances, the Steps-to-War theory does not find any evidence for the detrimental consequences of alliances.

Testing the Steps-to-War Theory with Heckman's Selection Model

In their 2017 study, Kenwick and Vasquez tested their Steps-to-War theory using logistic regression. However, this statistical model cannot adequately account for the escalation from minor disputes to full-scale war, which is a crucial aspect of the theory. Instead, they should have employed Heckman's selection model to more appropriately assess the spiral of international conflicts.

Kenwick and Vasquez's Steps-to-War theory suggests that states try to manage conflicts and crises by adopting *Realpolitik* policies such as arming themselves and forming security alliances. However, these policies can lead to a security dilemma for related states, causing increased fear and hostility and making war more likely. The theory proposes that after becoming involved in a low-intensity conflict, two states in a dyad will eventually reach a point where they precipitate toward war. This idea is critical to the Steps-to-War theory because it specifies a unique causal mechanism in which alliances push states into war. Therefore, this causal sequence requires further empirical testing, which Kenwick and Vasquez did not address in their logit analyses.

The previous section demonstrates that the formation of alliances does not have an aggravating effect on the initiation of war, which contradicts the predictions of the Steps-to-War theory. However, it is possible that states in newly formed alliances have fewer tensions to start with, which could explain why they experience fewer wars. States may be more reluctant to engage in militarized disputes when there is a higher risk of escalating to all-out war. This might lead to an underrepresentation of militarized disputes that would not escalate into war. Since wars can only occur following a militarized dispute, this results in an overrepresentation of militarized disputes with a lower probability of war outbreak (among states with defensive alliances) compared to disputes with a higher probability (among states without those alliances).

When we adjust for the base rate of militarized dispute occurrences related to defensive alliances, the model can detect that the cases with defensive alliances and no war outbreak should be down-weighted because of their higher probability of militarized dispute initiation. Cases with no defensive alliances (and more likely war outbreaks) should be up-weighted. This explanation is further supported by the negative error correlation term (ρ) discussed below.

By considering dispute dyads at a level below interstate war, this study isolates the effect of alliances on interstate war. This study employs a Heckman selection model to predict militarized disputes in the first stage and then analyze the impact of alliances on interstate war in the second stage, only considering dispute dyads.[14] For identification purposes, this study excludes one variable (i.e., "Similarity in alliance portfolios") from the war model because the selection equation (Model 1) should contain at least one variable that is not in the outcome equation (Model 2). This variable is likely related to the onset of militarized disputes in Model 1 since

[14] Other studies have also used a Heckman selection model for similar purposes (e.g., Reed 2000; Johnson and Leeds 2011; Piazza and Choi 2018), but their focus is not testing the Steps-to-*War* theory.

Table 3. Defense alliance and international conflict: dealing with an excessive number of irrelevant zeros.

Variable	Wars					
	Rare event logit			Politically relevant dyads		
	1816–2000 Model 1	1816–1945 Model 2	1946–2000 Model 3	1816–2000 Model 4	1816–1945 Model 5	1946–2000 Model 6
Target has a defense alliance from birth	−1.904***	no	−1.268*	−1.186*	no	−0.789
	(0.603)	variation	(0.681)	(0.612)	variation	(0.665)
Target has a defense alliance formed after birth	−1.286***	−0.422	−0.525	−1.148***	−0.668	−0.773
	(0.295)	(0.412)	(0.477)	(0.345)	(0.515)	(0.488)
Potential challenger has a relevant offensive alliance	0.980***	1.150***	no	0.919**	1.003**	no
	(0.344)	(0.401)	variation	(0.405)	(0.448)	variation
Potential challenger has a relevant neutrality pack	0.061	0.388	no	−1.016	−0.269	no
	(0.491)	(0.491)	variation	(0.636)	(0.634)	variation
Logged distance	−1.094***	−0.769***	.1 242***	−0.401***	−0.231	−0.570***
	(0.051)	(0.100)	(0.055)	(0.110)	(0.150)	(0.158)
Capabilities ratio	0.704**	1.153***	−0.349	−0.551	−0.286	−1.432*
	(0.326)	(0.406)	(0.517)	(0.402)	(0.460)	(0.766)
Joint democracy	−1.872**	no	−0.350	−1.669**	no	−0.600
	(0.760)	variation	(0.854)	(0.740)	variation	(0.719)
Similarity in alliance portfolios	−0.844	0.233	−1.401	−0.161	0.474	−1.129
	(0.539)	(0.700)	(1.567)	(0.659)	(1.076)	(1.270)
Constant	0.273	−3.411***	2.043	−2.554**	−5.025***	−0.273
	(0.659)	(0.968)	(1.632)	(1.057)	(1.402)	(1.977)
N	1,077,992	211,128	866,864	88,790	34,833	53,957

Notes: Numbers in parentheses are robust standard errors adjusted for clustering on directed dyads. *$P <. 10$; **$P <. 05$; ***$P <. 01$. t, t^2, and t^3 do not appear to economize space.

states are less likely to engage in a dispute when they agree on policy. However, once a dispute has been initiated, the variable becomes less relevant to the onset of war in Model 2 since states are already aware of a contentious issue between them. This satisfies the exclusion restriction of an instrumental variable.

There may be a concern that "Similarity in alliance portfolios" could influence the probability of war. States that share similar geopolitical interests, reflected in their alliances, might be less inclined to engage in all-out conflict with each other. To address this concern, I perform a robustness check by substituting "Similarity in alliance portfolios" with peace-year polynomials, which specifically relate to militarized disputes rather than wars.

Table 4 presents the estimates from the Heckman selection models that utilize "Similarity in alliance portfolios," while Table 5 provides estimates based on peace-year polynomials. The overall findings suggest that the Steps-to-War theory is not supported by empirical data when accounting for unobservable factors that influence the onset and escalation of disputes.

Models 1 and 2 are the first selection models covering the entire sample period from 1816 to 2000. In the first stage, this study tests for the influence of alliances on initiating any MID. In the second stage, the study examines whether MIDs that have occurred will escalate into war. These two models are constructed to more accurately test the prediction of Kenwick and Vasquez's Steps-to-War theory. Consistent with the Steps-to-War theory, this study finds the aggravating effects of defense pacts in Models 1, 3, and 5 in the first stage. However, the effects of the Steps-to-War variable on war in Models 2, 4, and 6 change drastically. In Model 2, "Target has a defense alliance formed after birth" is statistically different from zero, but its effect is benign. This implies that contrary to the Steps-to-War prediction, alliances introduced in a directed dyad even after birth significantly *decrease* the probability of war. The Steps-to-War variable still lacks empirical support when the sample data is divided into two sub-groups before and after World War II in Models 4 and 6.

In Table 4, the error correlation term (ρ) has a negative sign across the three models, indicating that the unmeasured variables that get dyads into a dispute hinder those disputes from escalating into war. The error correlation term for the second Heckman's selection model from 1816 to 1945 suggests that the two processes of MIDs and war relate to each other. This justifies the use of the second selection model for the onset of MIDs and their escalation to war. However, the error correlation term for the first and third selection models is insignificant, indicating that the two stages are independent.

Based on Table 4, figure 2 displays the substantive effects of alliances, which aligns with those in figure 1. While the deterrence theory variable substantively engenders the dampening effect on war, the Steps-to-War theory variable is not associated with the aggravating effect on war.

Table 5 presents a robustness test for Heckman selection models by excluding peace-year polynomials from the first stage. The estimated results are consistent with those in Table 4. The prediction of Steps-to-War is not supported; in fact, alliances formed in a directed dyad, even after their inception, significantly reduce the probability of war.

Based on the information in Table 4, but not Table 5, one may argue that there is no statistical (though not theoretical) justification for using a selection model to analyze the onset of disputes and their escalation to war. To address this issue, this study re-estimates the war initiation model. This approach allows us to assess whether hostilities escalate to the point of war for the periods 1816–2000 and 1946–2000.

Because standard logistic regression, which assumes complete data, can yield biased estimates when dealing with censored war data, I employ censored logistic regression. This method is specifically designed to manage this incompleteness and

Table 4. Heckman selection model of MIDs and wars: similarity in alliance portfolios excluded.

Variable	1816–2000		1816–1945		1946–2000	
	All MIDs Model 1	Wars Model 2	All MIDs Model 3	Wars Model 4	All MIDs Model 5	Wars Model 6
Target has a defense alliance from birth	−0.351***	−0.438*	−0.370*	−4.603***	−0.369***	−0.267
	(0.087)	(0.257)	(0.221)	(0.788)	(0.094)	(0.281)
Target has a defense alliance formed after birth	0.235***	−0.531***	0.590**	−0.145	0.221**	−0.429**
	(0.087)	(0.118)	(0.282)	(0.171)	(0.093)	(0.193)
Potential challenger has a relevant offensive alliance	0.264***	0.289	0.511***	−0.062	−0.074	−4.560***
	(0.038)	(0.180)	(0.045)	(0.209)	(0.080)	(0.374)
Potential challenger has a relevant neutrality pack	0.302***	−0.448**	0.126**	−0.143	0.417***	−4.767***
	(0.055)	(0.186)	(0.063)	(0.210)	(0.065)	(0.274)
Logged distance	−0.373***	0.013	−0.200***	−0.070	−0.493***	0.004
	(0.013)	(0.077)	(0.016)	(0.087)	(0.023)	(0.150)
Capabilities ratio	0.120***	−0.134	0.202***	0.059	0.062	−0.594**
	(0.043)	(0.153)	(0.064)	(0.177)	(0.055)	(0.278)
Joint democracy	−0.147***	−0.435*	−0.337***	−4.580***	−0.057	0.005
	(0.035)	(0.257)	(0.083)	(0.695)	(0.041)	(0.283)
Similarity in alliance portfolios	−0.475***		−0.098		−0.926***	
	(0.085)		(0.084)		(0.155)	
Constant	0.854***	−0.692**	−0.755***	0.498	2.024***	−0.471
	(0.124)	(0.284)	(0.138)	(0.535)	(0.283)	(0.471)
P	−0.270		−0.498**		−0.286	
	(0.167)		(0.207)		(0.277)	
N	1,077,992		211,128		866,864	
Uncersored N	2,354		872		1,482	

Notes: Numbers in parentheses are robust standard errors adjusted for clustering on directed dyads. *P <. 10; **P <. 05; ***P <. 01. t, t^2, and t^3 do not appear to economize space.

Table 5. Heckman selection model of MIDs and wars: peace-year polynomials excluded.

Variable	1816–2000		1816–1945		1946–2000	
	All MIDs Model 1	Wars Model 2	All MIDs Model 3	Wars Model 4	All MIDs Model 5	Wars Model 6
Target has a defense alliance from birth	−0.351***	−0.369	−0.370*	−4.584***	−0.368***	−0.108
	(0.087)	(0.256)	(0.221)	(0.659)	(0.094)	(0.279)
Target has a defense alliance formed after birth	0.236***	−0.593***	0.590**	−0.158	0.223**	−0.590***
	(0.087)	(0.122)	(0.282)	(0.172)	(0.093)	(0.177)
Potential challenger has a relevant offensive alliance	0.264***	0.210	0.511***	−0.080	−0.074	−4.512***
	(0.038)	(0.178)	(0.045)	(0.202)	(0.080)	(0.505)
Potential challenger has a relevant neutrality pack	0.302***	−0.490***	0.126**	−0.244	0.417***	−4.658***
	(0.055)	(0.182)	(0.063)	(0.207)	(0.065)	(0.344)
Logged distance	−0.373***	−0.002	−0.200***	−0.097	−0.493***	0.046
	(0.013)	(0.080)	(0.016)	(0.091)	(0.023)	(0.141)
Capabilities ratio	0.120***	−0.133	0.202***	0.085	0.062	−0.573**
	(0.043)	(0.152)	(0.064)	(0.182)	(0.055)	(0.271)
Joint democracy	−0.147***	−0.499**	−0.337***	−4.884***	−0.057	−0.094
	(0.035)	(0.255)	(0.083)	(0.582)	(0.041)	(0.255)
Similarity in alliance portfolios	−0.474***	−0.225	−0.096	−0.177	−0.925***	0.025
	(0.085)	(0.312)	(0.084)	(0.454)	(0.155)	(0.435)
Constant	0.853***	−0.377	−0.757***	0.727	2.023***	−0.541
	(0.124)	(0.474)	(0.137)	(0.700)	(0.283)	(0.741)
P	−0.308*		−0.469*		−0.453*	
	(0.166)		(0.256)		(0.261)	
N	1,077,992		211,128		866,864	
Uncersored *N*	2,354		872		1,482	

Notes: Numbers in parentheses are robust standard errors adjusted for clustering on directed dyads. *P <. 10; **P <. 05; ***P <. 01.

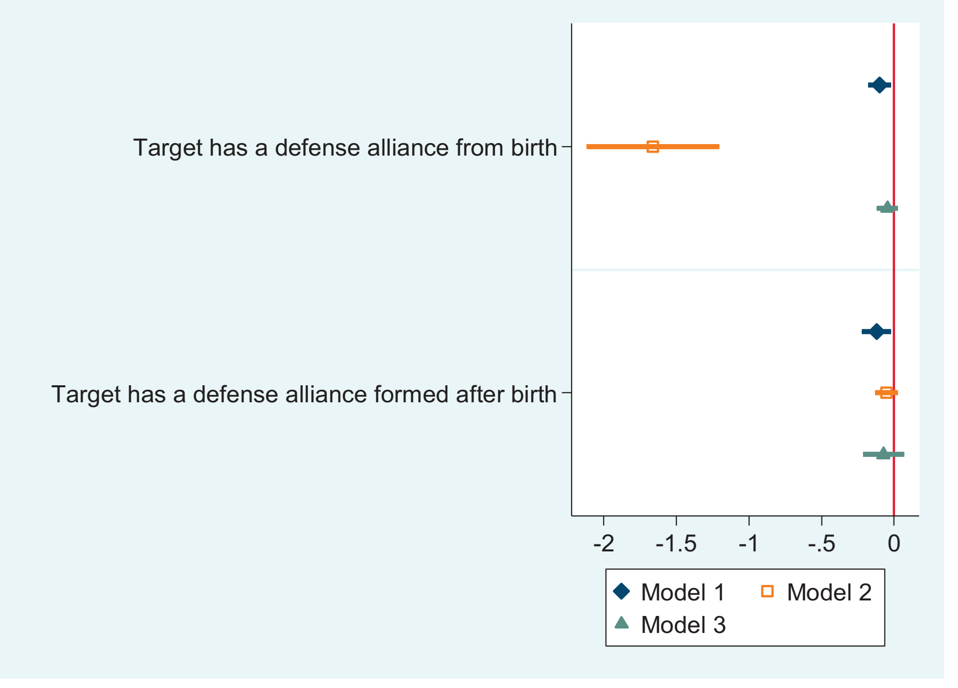

Figure 2. Average substantive effects

provide more accurate estimates. Censored logistic regression estimates the probability of war initiation while accounting for the fact that some countries may be observed for shorter durations than others, resulting in unobserved instances of war initiation.

Table 6 presents the estimates from the censored logistic regression models concerning war. It is essential to note that the overall results of significance tests in the censored logit models in Table 6 closely resemble those in the selection models presented in Table 4. In Table 6, the alliance variables are statistically significant in Models 1 and 2, with one exception, and their coefficients are negative. These findings confirm the predictions of deterrence theory while contradicting those of the Steps-to-War theory.

Concluding Remarks

The effectiveness of alliances in deterring wars has been a topic of considerable debate. In a prominent journal, scholars in the areas of deterrence and Steps-to-War presented differing views, which many appreciated as an effort to clarify whether alliances have a deterring or exacerbating effect on conflicts. This scholarly exchange was seen as a means to advance scientific knowledge. This research note commends Kenwick and Vasquez (2017) for adopting Leeds and Johnson's (2011) logit model, an attempt to demonstrate that their findings remain valid even when using the most commonly employed estimation technique in alliance politics literature. Their decision to enhance comparability was in response to criticism from Leeds and Johnson (2017, 335), who stated that "we do not believe that [Kenwick, Vasquez, and Powers' (2015)] research design is appropriate for testing the theoretical argument we have made in our past work." The criticism from Leeds and Johnson was persuasive, as Kenwick, Vasquez, and Powers' (2015) matching analysis relied on a drastically reduced sample size, representing only about 4 percent of the

Table 6. The effect of defense alliance on the escalation of MIDs to war.

Variable	Censored logit	
	1816–2000 Model 1	1946–2000 Model 2
Target has a defense alliance from birth	−1.428**	−0.942
	(0.598)	(0.677)
Target has a defense alliance formed after birth	−1.234***	−0.996**
	(0.311)	(0.461)
Potential challenger has a relevant offensive alliance	0.603	(No variation)
	(0.379)	
Potential challenger has a relevant neutrality pack	−0.771	(No variation)
	(0.469)	
Logged distance	−0.206**	−0.347**
	(0.094)	(0.163)
Capabilities ratio	−0.247	−1.564**
	(0.334)	(0.671)
Joint democracy	−1.183*	0.047
	(0.682)	(0.686)
Similarity in alliance portfolios	−0.886	−1.155
	(0.667)	(1.181)
Constant	−0.084	1.361
	(0.987)	(1.631)
N	2,354	1,482

Notes: Numbers in parentheses are robust standard errors adjusted for clustering on directed dyads. $*P < .10$; $**P < .05$; $***P < .01$. t, t^2, and t^3 do not appear to economize space.

sample size used by Johnson and Leeds (2011). Despite their efforts to align their statistical analyses with those of Leeds and Johnson, notable differences remained.

In their 2017 study, Kenwick and Vasquez overlooked two crucial tests. First, their model does not explore how alliances affect war, despite their Steps-to-War theory primarily addressing war initiation. Second, they fail to devise a statistical model that suitably captures the Steps-to-War theory, which emphasizes that conflicts escalate from low-intensity disputes to war. In short, while Kenwick and Vasquez's Steps-to-War theory focuses on war, their empirical models neglect to test war itself. After addressing these issues, this research note concludes that the Steps-to-War theory has limited explanatory power in predicting the relationship between defensive alliances and war (for insights on the effectiveness of the Steps-to-War theory, see also Vasquez 1993; Senese and Vasquez 2008; Kenwick, Vasquez, and Powers 2015). This new finding directly contradicts Kenwick and Vasquez's (2017, 331) assertion in the Scholarly Dialogue: "Neither do Leeds and Johnson (2017) address our findings on war. . . Our article, however, is not mainly a critique of their work but an analysis of deterrence. From a policy perspective, what is the point of deterring MIDs if the same logic cannot prevent war?"

Although Vasquez and his colleagues have made commendable efforts to develop the Steps-to-War theory, it still fails to clearly explain how militarized disputes escalate into war. As in Kenwick and Vasquez's (2017) study, the theory lacks a detailed mechanism for understanding the transition from disputes to armed conflict. This research note demonstrates that the Steps-to-War theory does not provide supporting evidence for the aggravating effect of alliances on war outbreaks. The implication is that the validity of the theory is in question and requires serious revisions. In this regard, Owsiak (2017, 20) aptly notes: "It seems that alliance relationships are complex. More research is needed to understand which alliance relationships promote peace, which promote war, and whether their effects remain constant over time."

One may wonder why the Steps-to-War theory predicts the likelihood of militarized disputes but not war. This research note suggests that the Steps-to-War variable (i.e., "Target has a defense alliance formed after birth") appears to be subject to a ceiling effect. Although the theory predicts an increased probability of war, data analyses do not produce supporting evidence for this effect.[15] Alliance formation may lead to militarized disputes if the challenger wishes to test the waters. For the challenger, low-intensity conflicts can serve as a means to gather information about the target state's response to a plan or to assess the strength of the target state's bond with its new military ally.

However, alliance formation is unlikely to provoke war because the challenger understands that the risks are too high. In a full-scale war, the challenger would have to fight at least two opponents: the target state and its ally. Consequently, the potential benefits may not be significant enough for the challenger to justify the costs of military invasion when the target has a security ally to fall back on (Johnson and Leeds 2011; Leeds and Johnson 2017).

In sum, when analyzing the prospect of war, Kenwick and Vasquez's Steps-to-War theory proved inadequate, while Leeds and Johnson's alliance theory held up. These findings suggest that, contrary to Kenwick and Vasquez's misguided recommendations, political leaders should seek new alliance partners to safeguard their nations and promote interstate peace. The reasoning is clear: statistical evidence consistently supports Johnson and Leeds' (2011) deterrence theory, demonstrating that alliances do deter *war*.

Supplementary material

Supplementary material is available at *Foreign Policy Analysis* online.

References

Bak, Daehee. 2018. "Alliance Proximity and Effectiveness of Extended Deterrence." *International Interactions* 44: 107–31. https://doi.org/10.1080/03050629.2017.1320995

Benson, Brett, and Bradley Smith 2023. "Commitment Problems in Alliance Formation." *American Journal of Political Science* 67: 1012–25. https://doi.org/10.1111/ajps.12693

Chen, Frederick. 2021. "Extended Dependence: Trade, Alliances, and Peace." *The Journal of Politics* 83: 246–59. https://doi.org/10.1086/709149

Choi, Seung-Whan. 2011. "Re-evaluating Capitalist and Democratic Peace Models." *International Studies Quarterly* 55: 759–69. https://doi.org/10.1111/j.1468-2478.2011.00674.x

Choi, Seung-Whan. 2013, "The Democratic Peace through an Interaction of Domestic Institutions and Norms." *Armed Forces & Society* 39: 255–83.

Choi, Seung-Whan. 2022. "Nationalism and Withdrawals from Intergovernmental Organizations." *The Review of International Organizations* 17: 205–15. https://doi.org/10.1007/s11558-021-09417-1

Choi, Seung-Whan. 2024. "Can Trade and Security Alliance help reduce Interstate War?" *PLoS One* 19: e0304482. https://doi.org/10.1371/journal.pone.0304482

Choi, Seung-Whan, and Patrick James. 2025. "Six Alternatives to War, One Solution for Peace." *The Chinese Journal of International Politics* 18: 1–34. https://doi.org/10.1093/cjip/poae023

Clare, Joe. 2013. "The Deterrent Value of Democratic Allies." *International Studies Quarterly* 57: 545–55https://doi.org/10.1111/isqu.12012

Dafoe, Allan, John Oneal, and Bruce Russett. 2013. "The Democratic Peace." *International Studies Quarterly* 57: 201–14. https://doi.org/10.1111/isqu.12055

Fearon, James. 2002. "Selection Effects and Deterrence." *International Interactions* 28: 5–29. https://doi.org/10.1080/03050620210390

[15] Other theories, such as the democratic peace, are less vulnerable to a ceiling effect (e.g., Choi 2011; Choi and James 2025).

Huth, Paul, and Bruce Russett. 1984. "What Makes Deterrence Work?" *World Politics* 36: 496–526. https://doi.org/10.2307/2010184

Johnson, Jesse, Brett Ashley Leeds, and Ahra Wu. 2015. "Capability, Credibility, and Extended General Deterrence." *International Interactions* 41: 309–36. https://doi.org/10.1080/03050629.2015.982115

Johnson, Jesse, and Brett Ashley Leeds. 2011. "Defense Pacts." *Foreign Policy Analysis* 7: 45–5. https://doi.org/10.1111/j.1743-8594.2010.00122.x

Kenwick, Michael, and John Vasquez. 2017. "Defense Pacts and Deterrence." *The Journal of Politics* 79: 329–34. https://doi.org/10.1086/686700

Kenwick, Michael, John Vasquez, and Matthew Powers. 2015. "Do Alliances Really Deter?" *The Journal of Politics* 77: 943–54. https://doi.org/10.1086/681958

Lee, Soyoung. 2023. "The Strong, the Weak, and the Honored: Examining the Decline in Honored Alliances Post-1945." *International Interactions* 49: 114–31. https://doi.org/10.1080/03050629.2023.2135513

Leeds, Brett Ashley. 2003. "Do Alliances deter Aggression?" *American Journal of Political Science* 47: 427–39. https://doi.org/10.1111/1540-5907.00031

Leeds, Brett Ashley. 2015. "Why Do States Sign Alliances?" In *Emerging Trends in the Social and Behavioral Sciences*, edited by Robert Scott and Stephen Kosslyn, 1–14. Hoboken: Wiley. https://doi.org/10.1002/9781118900772.

Leeds, Brett Ashley, and Jesse Johnson. 2017. "Theory, Data, and Deterrence." *The Journal of Politics* 79: 335–40. https://doi.org/10.1086/687285

Levy, Jack. 1981. "Alliance Formation and War Behavior." *Journal of Conflict Resolution* 25: 581–613. https://doi.org/10.1177/002200278102500402

Morrow, James. 2000. "Alliances: Why Write Them Down?" *Annual Review of Political Science* 3: 63–83. https://doi.org/10.1146/annurev.polisci.3.1.63

Owsiak, Andrew. 2017. "The Steps to War: Theory and Evidence." In *The Oxford Research Encyclopedia of International Studies*, edited by N. Sandal, 1–30.New York: Oxford University Press.

Piazza, James, and Seung-Whan Choi. 2018. "International Military Interventions and Transnational Terrorist Backlash." *International Studies Quarterly* 62: 686–95. https://doi.org/10.1093/isq/sqy026

Pressman, Jeremy. 2008. *Warring Friends: Alliance Restraint in International Politics.* Ithaca, NY: Cornell University Press.

Quackenbush, Stephen. 2010. "General Deterrence and International Conflict: Testing Perfect Deterrence Theory." *International Interactions* 36: 60–85. https://doi.org/10.1080/03050620903554069

Reed, William. 2000. "A Unified Statistical Model of Conflict Onset and Escalation." *American Journal of Political Science* 44: 84–93. https://doi.org/10.2307/2669294

Rynning, Sten, and Olivier Schmitt. 2018. "Alliances." In *The Oxford Handbook of International Security*, edited by Alexandra Gheciu and William Wohlforth, 653–67. Oxford: Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198777854.001.0001

Senese, Paul, and John Vasquez. 2008. *The Steps to War.* Princeton, NJ: Princeton University Press.

Smith, Alastair. 1995. "Alliance Formation and War." *International Studies Quarterly* 39: 405–25. https://doi.org/10.2307/2600800

Snyder, Gleen. 1997. *Alliance Politics.* Ithaca, NJ: Cornell University Press.

Vasquez, John. 1993. *The War Puzzle.* Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511583483

Wright, Thorin, and Toby Rider. 2014. "Disputed Territory, Defensive Alliances, and Conflict Initiation." *Conflict Management and Peace Science* 31: 119–44. https://doi.org/10.1177/0738894213503440

Yuen, Amy. 2009. "Target Concessions in the Shadow of Intervention." *Journal of Conflict Resolution* 53: 745–73. https://doi.org/10.1177/0022002709339046

Choi, Seung-Whan (2025) Do Alliances Deter War?. *Foreign Policy Analysis*, https://doi.org/10.1093/fpa/oraf027

Appendix 1 A Correlation Matrix of Variables: MIDs

	MIDs	Birth alliance	After-birth alliance	Offensive alliance	Neutrality pack	Logged distance	Capabilities ratio	Joint democracy	Similarity
MIDs	1.000								
Birth alliance	-0.016	1.000							
After-birth alliance	0.004	-0.198	1.000						
Offensive alliance	0.017	-0.054	0.038	1.000					
Neutrality pack	0.014	-0.033	0.013	0.052	1.000				
Logged distance	-0.069	0.076	-0.037	-0.066	0.022	1.000			
Capabilities ratio	0.010	-0.178	0.000	0.083	0.066	0.000	1.000		
Joint democracy	-0.004	-0.037	0.051	-0.051	0.007	-0.020	0.000	1.000	
Similarity	0.007	-0.143	-0.274	-0.078	-0.020	-0.254	0.000	-0.005	1.000

Appendix 2 A Correlation Matrix of Variables: War

	War	Birth alliance	After-birth alliance	Offensive alliance	Neutrality pack	Logged distance	Capabilities ratio	Joint democracy	Similarity
War	1.000								
Birth alliance	-0.004	1.000							
After-birth alliance	-0.004	-0.198	1.000						
Offensive alliance	0.005	-0.054	0.038	1.000					
Neutrality pack	0.000	-0.033	0.013	0.052	1.000				
Logged distance	-0.017	0.076	-0.037	-0.066	0.022	1.000			
Capabilities ratio	0.003	-0.178	0.000	0.083	0.066	0.000	1.000		
Joint democracy	-0.003	-0.037	0.051	-0.051	0.007	-0.020	0.000	1.000	
Similarity	0.002	-0.143	-0.274	-0.078	-0.020	-0.254	0.000	-0.005	1.000